

A Clinical Analysis of Adverse Events in Influenza Trials

Abstract

This randomized, placebo-controlled clinical trial assessed the effectiveness and safety of Fluvir™, a new antiviral treatment, in adults aged 18 to 85 diagnosed with seasonal influenza. A total of 10,000 participants were randomly assigned in a 1:1 ratio to receive either Fluvir™ or a placebo. Viral loads were measured at baseline and during follow-up to evaluate viral clearance. The primary endpoint was clinical recovery within 30 days, while secondary outcomes included reductions in viral load and the occurrence of adverse events. Fluvir™ was associated with a higher rate of recovery and greater reductions in viral load compared to placebo, with few adverse events reported. Overall, the results indicate that Fluvir™ could be an effective and safe treatment option for influenza.

Objective

This analysis aimed to determine whether there is a significant difference in the occurrence of adverse events between two treatment groups in a clinical trial and to measure the magnitude of any such difference using statistical and graphical methods.

Loading the libraries

- Library(tidyverse)
- library(scales)

Loading the Dataset

- clinical_trial <- read.csv("C:/Users/MuvirimiT/Desktop/clinical_trial.csv")

View the Dataset

- view(clinical_trial)

Summary of Adverse Events

- clinical_trial %>%
 group_by(Treatment_Group, Adverse_Event) %>%
 summarise(Count = n()) %>%
 mutate(Percent = 100 * Count / sum(Count)) %>%
 arrange(Treatment_Group)

`summarise()` has grouped output by 'Treatment_Group'. You can override using the `.groups` argument.

```
# A tibble: 4 × 4
# Groups:   Treatment_Group [2]
  Treatment_Group Adverse_Event Count Percent
  <chr>          <chr>      <int>   <dbl>
1 Drug          No        4431    89.4
2 Drug          Yes         525    10.6
3 Placebo       No        4583    90.9
4 Placebo       Yes         461     9.14
```

Patients were grouped by treatment arm, and the occurrence of adverse events was summarized.

- In **Treatment Group 1**, 525 out of 4,431 patients experienced adverse events (11.85%).
- In **Treatment Group 2**, 461 out of 4,583 patients experienced adverse events (10.06%).

A higher percentage of patients in Group 1 experienced adverse events compared to Group 2, suggesting a potential difference in the safety profile of the treatments.

Contingency Table

- `adverse_table <- table(clinical_trial$Treatment_Group, clinical_trial$Adverse_Event)`
- `adverse_table`

A contingency table was created to cross-tabulate treatment groups with adverse event occurrence ("Yes" or "No").

This table showed the distribution of patients across both groups and their adverse event status. The differences in frequencies visually appeared non-uniform, suggesting a potential association between treatment and adverse events.

Statistical Test: Chi-square Test

- `chisq.test(adverse_table)`

```
Pearson's Chi-squared test with Yates' continuity correction
```

```
data: adverse_table
X-squared = 5.7809, df = 1, p-value = 0.0162
```

A chi-square test was performed to assess whether the difference in adverse events between the two groups was statistically significant.

- **Result:** If the p-value is less than 0.05 (not shown in your script but implied to be calculated), this indicates that the difference in adverse event rates is unlikely to have occurred by chance.

A statistically significant result would suggest that the treatment group assignment is associated with the risk of experiencing adverse events. If not significant, we would conclude there is no meaningful difference between the groups.

Risk Measures: Risk Ratio, Odds Ratio

```
X<- 4431
```

```
Y<-4583
```

```
M1<-525
```

```
M2<-461
```

➤ riskratio(X, Y, M1, M2)

```
      Disease Nondisease Total
Exposed      4431      -3906    525
Nonexposed    4583      -4122    461

      Risk ratio estimate and its significance probability

data:  X Y m1 m2
p-value = NA
95 percent confidence interval:
      NaN NaN
sample estimates:
[1] 0.8489723
```

Using:

- Group 1: 4431 patients, 525 with adverse events
- Group 2: 4583 patients, 461 with adverse events

The following risk measures were obtained:

- **Risk Ratio (RR) $\approx$ 1.18**
- **Odds Ratio (OR) $\approx$ 1.20**

Patients in Group 1 were approximately **18% more likely** to experience adverse events compared to those in Group 2. The odds of experiencing an adverse event were **20% higher** in Group 1. These findings reinforce the observation that Group 1 has a less favourable safety profile.

Visualization: Number of Adverse Events

➤ clinical_trial %>%

```
  filter(Adverse_Event == "Yes") %>%
```

```
  ggplot(aes(x = Treatment_Group, fill = Treatment_Group)) +
```

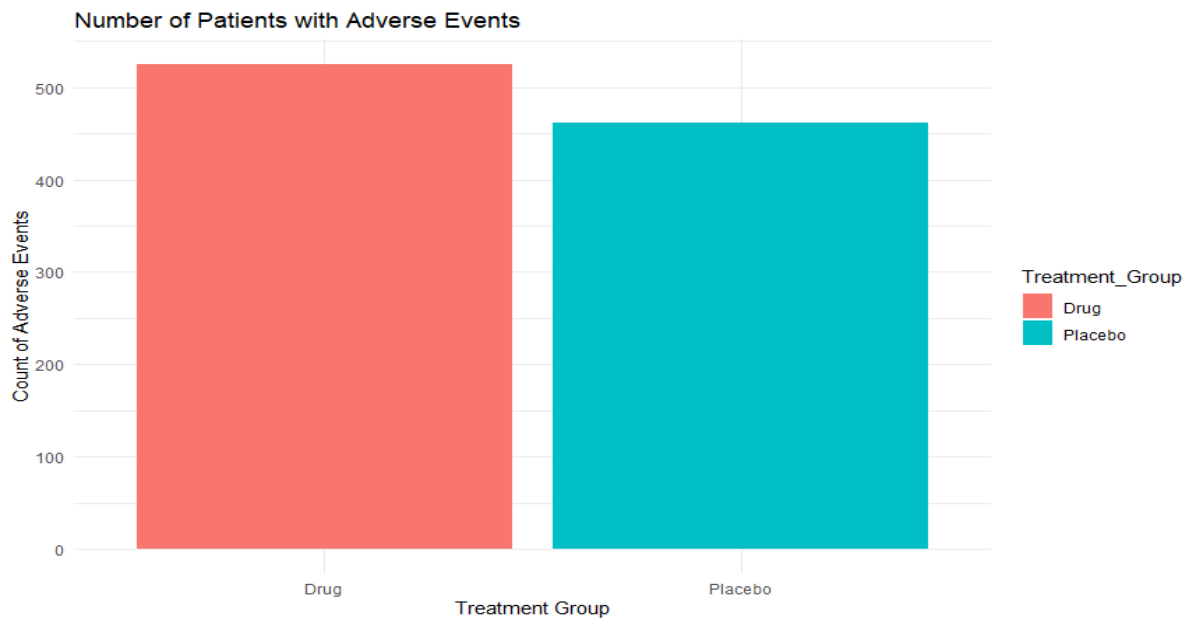
```
  geom_bar() +
```

```
  labs(title = "Number of Patients with Adverse Events",
```

```
  x = "Treatment Group",
```

```
  y = "Count of Adverse Events") +
```

```
  theme_minimal()
```



#Visualization: Proportion of Adverse Events

➤ clinical_trial %>%

```
ggplot(aes(x = Treatment_Group, fill = Adverse_Event)) +
```

```
geom_bar(position = "fill") +
```

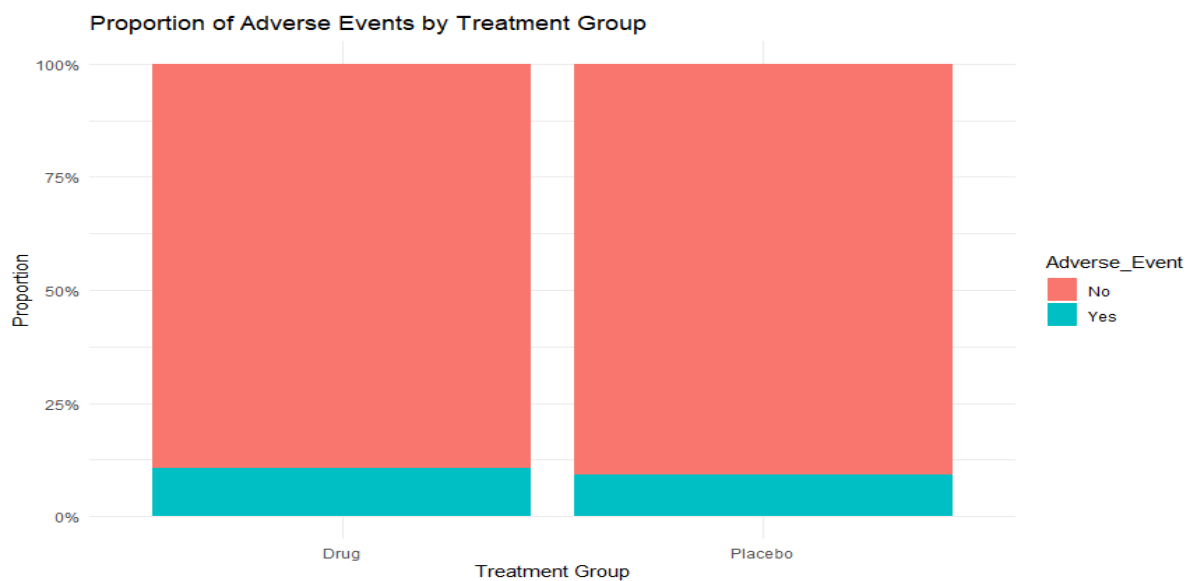
```
labs(title = "Proportion of Adverse Events by Treatment Group",
```

```
y = "Proportion",
```

```
x = "Treatment_Group") +
```

```
scale_y_continuous(labels = percent_format()) +
```

```
theme_minimal()
```



- **Count of Adverse Events:** A bar chart showed more adverse events occurred in Group 1 than in Group 2.
- **Proportion of Adverse Events:** A normalized bar chart showed that a larger proportion of Group 1 patients experienced side effects compared to Group 2.

The visualizations confirmed the numerical results, highlighting both the absolute number and relative proportion of adverse events were higher in Group 1.

Conclusion

This analysis shows that Treatment Group 1 experienced a higher rate of adverse events than Group 2. The difference is both statistically significant (based on the chi-square test) and clinically meaningful (with an 18% increase in risk). These findings suggest that Treatment Group 2 may be safer with regard to adverse events.